

William Rule

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UC Berkeley Applied Mathematics | Statistics Minor | Python | Data Analysis | Numerical Methods | Computational Research

EDUCATION

University of California, Berkeley - B.A. Applied Mathematics; Planned Minor in Statistics Planned/In-Progress Coursework: Math 110 Linear Algebra, Stat 134 Probability, Math 128A Numerical Analysis, CS 61A, Optimization	Expected May 2028
Saddleback College - Mathematics (Honors) GPA: 4.00/4.00 Completed Coursework: Calculus I-III, Differential Equations, Linear Algebra, Discrete Mathematics	Jan 2025 - May 2026

TECHNICAL SKILLS

Programming: Python, NumPy, pandas, SciPy, matplotlib, Git/GitHub, MATLAB | **Data & Modeling:** numerical methods, simulation, backtesting, data visualization

RESEARCH EXPERIENCE

Research Assistant - Economics Python/oTree Developer Chapman University - Built Python/oTree platform for a faculty-led public-goods experiment; implemented workflows, payoff logic, UI, data export, and bot tests. - Supported reproducible data collection for 568 participants across a 2x2 experimental design.	2025 - Present
Visualizing Tokenized Carbon Credit Markets Python Data Analysis - Analyzed 50,000+ Klima DAO transaction records using Python to study liquidity, transparency, and usability patterns in blockchain-based markets. - Presented at the UC Berkeley Bay Honors Research Symposium; published a 19-page paper and code; ranked as a Top-10 recent download in two SSRN eJournals.	2025

TECHNICAL & RESEARCH PROJECTS

Binomial Pricing Model & Black-Scholes Convergence Python, NumPy - Implemented an N-step numerical model in Python to study option-pricing convergence against the Black-Scholes as a benchmark. - Built no-arbitrage checks and backward-induction logic; visualized pricing error across increasing step counts. - Documented the replication intuition connecting discrete-time numerical approximation to continuous-time model behavior.	2026
Black-Scholes Taylor-Series Approximation Python, NumPy, SciPy - Derived 3rd/4th-order Taylor approximations for near-ATM European call prices; benchmarked estimated prices and Greeks against closed-form outputs. - Evaluated accuracy/runtime tradeoffs across tested parameter ranges using vectorized implementation.	2025
SPY Moving-Average Strategy Backtest Python, pandas - Built vectorized time-series backtester across 200+ moving-average pairs on 5 years of SPY data; computed CAGR, Sharpe, and max drawdown. - Produced heatmaps and results tables to identify parameter regimes with stronger risk-adjusted performance than buy-and-hold.	2025

TECHNICAL EXPERIENCE

Math Tutor, Saddleback College Learning Resource Center Tutored Calculus III and Differential Equations one-on-one and in groups for 175+ hours, emphasizing conceptual setup, derivations, and exam prep.	Aug 2025 - Present
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LEADERSHIP & OPERATIONS

Saddleback Student Government - Budget and Finance Committee, Director Oversee approximately \$433K annual budget; built ledgers/tracking systems; trained 9 officers; increased form completion by 60%.	Feb 2025 - Present
Saddleback Investing Club, Treasurer / Board Member Managed funding/allocation workflow; helped grow membership from 4 to 45 and co-ran technical analysis workshops.	2025 - Present
Nurture Aid Vision (NAV) - 501(c)(3) Nonprofit, Co-Founder, Finance/Ops Built website and QuickBooks system; raised \$10,000+ and delivered 1,000+ supply kits to 11 schools.	2024 - Present

AWARDS & CERTIFICATIONS

FINRA Securities Industry Essentials (SIE); Honors Research Scholar (Bay Honors Symposium, UC Berkeley); SSRN Top-Downloaded Paper; DataCamp certificates (Python, pandas, time series)